

Infantile Cerebellar-Retinal Degeneration (ICRD): A Comprehensive Disease Characterization Report

Disease: Infantile Cerebellar-Retinal Degeneration **MONDO ID:** MONDO:0013802 · **OMIM:** #614559 · **Orphanet:** ORPHA:314629 **Category:** Mendelian (autosomal recessive neurometabolic disorder) **Causal gene:** *ACO2* (mitochondrial aconitase; HGNC:118; UniProt Q99798)

Summary

Infantile Cerebellar-Retinal Degeneration (ICRD) is an ultra-rare, autosomal recessive, infantile-onset neurometabolic and neurodegenerative disorder caused by **biallelic loss-of-function variants in *ACO2***, the nuclear-encoded gene for mitochondrial aconitase — the second enzyme of the tricarboxylic acid (TCA)/Krebs cycle, which interconverts citrate and isocitrate via a catalytic [4Fe-4S] iron–sulfur cluster. Disease was first defined in 2012 by homozygosity mapping and whole-exome sequencing in two families sharing a homozygous p.Ser112Arg (c.336C>G) founder mutation, with severely reduced aconitase activity in patient lymphoblasts and failure of the mutant human enzyme to rescue a yeast aconitase-null strain ([PMID: 22405087](#)).

Clinically, ICRD presents in early infancy (typically 2–6 months) with severe truncal hypotonia, truncal ataxia, evolving microcephaly, variable seizures, and progressive ophthalmologic disease — esotropia and optic atrophy followed by retinal dystrophy. Brain MRI is usually normal in the first months but develops progressive global atrophy that is predominantly cerebellar. Most patients remain nonambulatory and nonverbal, though the phenotype is a genuine spectrum: at the severe end, infantile death; at the mild end, **isolated optic atrophy 9 (OPA9)** with preserved cognition and normal lifespan. This allelic-series relationship (severe ICRD ↔ mild OPA9) is a defining feature of *ACO2* disease.

Mechanistically, aconitase deficiency produces a coherent causal chain: (1) mutation damages the catalytic [4Fe-4S] cluster or substrate-binding residues → (2) loss of citrate→isocitrate conversion → (3) TCA-cycle/bioenergetic failure with secondary mitochondrial DNA depletion and deficient respiration → (4) toxic citrate accumulation that activates the integrated stress response (ISR) → (5) caspase-3-mediated apoptosis and reduced histone-acetylation-driven autophagy suppression → (6) preferential death of high-energy-demand neurons (cerebellar, retinal, optic) → (7) the clinical phenotype. Management is currently supportive; anaplerotic **triheptanoïn** (an odd-chain C7 triglyceride that refills TCA intermediates downstream of the aconitase block) has been trialed in two brothers as a rational but still-unproven disease-directed therapy.

Section 1 — Disease Information

Overview. ICRD is a Mendelian mitochondrial-adjacent (nuclear gene, mitochondrial enzyme) neurodegenerative disease of infancy defined by the triad of progressive cerebellar degeneration, retinal/optic degeneration, and global developmental delay/regression. It is a distinct clinical entity within the broader group of *ACO2*-related disorders.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0013802
OMIM	#614559 (Infantile cerebellar-retinal degeneration)
Orphanet	ORPHA:314629
Gene (HGNC)	<i>ACO2</i> , HGNC:118
Gene OMIM	100850
UniProt (protein)	Q99798 (Aconitate hydratase, mitochondrial)
Ensembl	ENSG00000100412

Synonyms / alternative names. ICRD; *ACO2*-related infantile cerebellar-retinal degeneration; mitochondrial aconitase deficiency; aconitase 2 (*ACO2*) deficiency. The allelic milder disorder is **Optic Atrophy 9 (OPA9)**.

Source of information. The knowledge base entry is derived from **aggregated disease-level resources** (OMIM, Orphanet) and from **individual-patient primary literature** — small case series and cohorts (the largest being 16 patients, [PMID: 30689204](#)) and single case reports. No EHR-scale population data exist given disease rarity.

Section 2 — Etiology

Primary cause (genetic). ICRD is caused exclusively by **biallelic (homozygous or compound heterozygous) pathogenic variants in ACO2**. There is no environmental or infectious etiology. The founding study established causality through homozygosity mapping plus WES, biochemical demonstration of severely reduced aconitase activity, and a yeast complementation assay in which mutant human ACO2 failed to rescue an *ACO1*-deletion strain ([PMID: 22405087](#)). A 16-patient multicenter cohort confirmed biallelic pathogenic ACO2 variants as the recurrent cause ([PMID: 30689204](#)).

Genetic risk factors. The causal variants themselves are the risk factor. A recurrent **founder allele c.336C>G (p.Ser112Arg)** appeared in 10 of 16 patients in the largest cohort ([PMID: 30689204](#)). ACO2 is highly constrained against variation in gnomAD (pLI 0.97; missense-Z 4.58; LoF-Z 4.94), consistent with an essential gene whose complete loss is not tolerated (Finding F004).

Environmental / protective factors, gene–environment interactions. None established. As a fully penetrant Mendelian recessive disorder, there are no recognized environmental risk factors, protective alleles, or GxE interactions. Consanguinity elevates recurrence risk of the recessive genotype but is not a disease cause per se.

Section 3 — Phenotypes

Phenotypes are drawn chiefly from the 16-patient cohort ([PMID: 30689204](#)) and the founding cohort ([PMID: 22405087](#)). Onset is neonatal-to-early-infantile; course is **progressive**.

Phenotype	Type	HPO term	Onset / severity / frequency
Truncal hypotonia	Clinical sign	HP:0008936	Early infancy; severe; most patients
Truncal ataxia	Clinical sign	HP:0002078	Infancy; severe; most patients
Optic atrophy	Physical/ ophthalmologic	HP:0000648	Infancy→childhood; dominant feature
Esotropia	Clinical sign	HP:0000565	Infancy; most dominant ocular sign
Retinal dystrophy/ degeneration	Physical	HP:0000556	Later than optic atrophy; progressive
Seizures	Clinical sign	HP:0001250	Variable; subset of patients
Microcephaly (evolving/acquired)	Physical	HP:0000252 / HP:0005484	Postnatal, progressive
Global developmental delay	Behavioral/ cognitive	HP:0001263	Infancy; severe; most patients
Intellectual disability	Cognitive	HP:0001249	Severe–profound (variable)
Cerebellar atrophy	Imaging	HP:0001272	Develops after normal early MRI
Cerebral (cortical) atrophy	Imaging	HP:0002059	Progressive
Absent speech/ language	Behavioral	HP:0001344	Most remain nonverbal
Peripheral neuropathy	Clinical sign	HP:0009830	Reported in longer-surviving cases (P 28545339)
Pigmentary retinopathy	Physical	HP:0000580	Reported in moderate/older cases

Characteristics summary (F008): Most patients present in early infancy with "severe truncal hypotonia, truncal ataxia, variable seizures, evolving microcephaly, and ophthalmological abnormalities of which the most dominant are esotropia and optic atrophy with later development of retinal dystrophy" (PMID: 30689204). "Brain magnetic resonance imaging (MRI) is typically normal within the first months but global atrophy gradually develops affecting predominantly the cerebellum" (same source). The founding study documented onset at 2–6 months with survival up to 18 years (PMID: 22405087).

Severity spectrum. Severe (infantile death) → moderate (increased survival with partly preserved cognition; a patient able to "speak full sentences and follow commands," PMID: 28545339) → mild isolated optic atrophy (OPA9).

Quality-of-life impact. Profound in classic ICRD: most patients are nonambulatory, nonverbal, visually impaired, and fully dependent for daily activities, with high caregiver burden. No formal EQ-5D/SF-36 instruments have been applied given rarity.

Section 4 — Genetic / Molecular Information

Causal gene. *ACO2* (aconitase 2, mitochondrial), HGNC:118, chr22q13.2 (GRCh38 chr22:41,447,830–41,529,273), Ensembl ENSG00000100412, protein UniProt Q99798 (780 aa).

Representative pathogenic variants.

Variant (protein)	cDNA	Type	Significance	Note / PMID
p.Ser112Arg	c.336C>G	Missense (founder)	Pathogenic	First ICRD variant; 10/16 cohort; 22405087, 30689204
p.Cys448Ser	—	Missense	Pathogenic	Removes [4Fe-4S] cluster ligand; 32713659
p.Met393Ile	—	Missense	Likely pathogenic	Adjacent to Cys385 cluster ligand; 32713659

Variant classification & type. Variants are predominantly **missense**, classified pathogenic/likely-pathogenic by ACMG/AMP criteria supported by functional enzyme assays. Nonsense/frameshift alleles also occur. Given strong LoF constraint (gnomAD LoF-Z 4.94, pLI 0.97), complete biallelic null genotypes may be embryonic-lethal, which is consistent with the missense-heavy spectrum observed in survivors.

Allele frequency. Pathogenic alleles are ultra-rare in gnomAD; overall *ACO2* is strongly depleted of both missense and LoF variation (Finding F004).

Origin & functional consequence. All disease alleles are **germline**; there is no somatic/cancer role. The molecular consequence is **loss of function** — reduced aconitase catalytic activity (patient fibroblasts <20% of control; [PMID: 26992325](#)), reduced cellular respiration, and secondary mitochondrial DNA depletion, all rescued by reintroducing wild-type *ACO2* (same study).

Structural basis (F005). UniProt Q99798 annotates an N-terminal mitochondrial transit peptide (aa 1–27), substrate-binding residues (99, 192–194, 474, 479, 607, 670–671), and three **[4Fe-4S] cluster-coordinating cysteines at 385, 448, 451**. p.Cys448Ser eliminates one cluster ligand; p.Met393Ile sits immediately adjacent to Cys385 — providing a direct structural explanation for enzyme failure.

Modifier genes / epigenetics / chromosomal abnormalities. No specific modifier genes are established; residual aconitase activity (allele-dependent) is the principal severity determinant. Notably, **LONP1** protease regulates *ACO2* turnover/stability ([PMID: 42302976](#)), a plausible modifier axis. Downstream epigenetic dysregulation (histone acetylation) is a consequence, not a cause (see Section 6). No recurrent large chromosomal abnormalities are associated with ICRD.

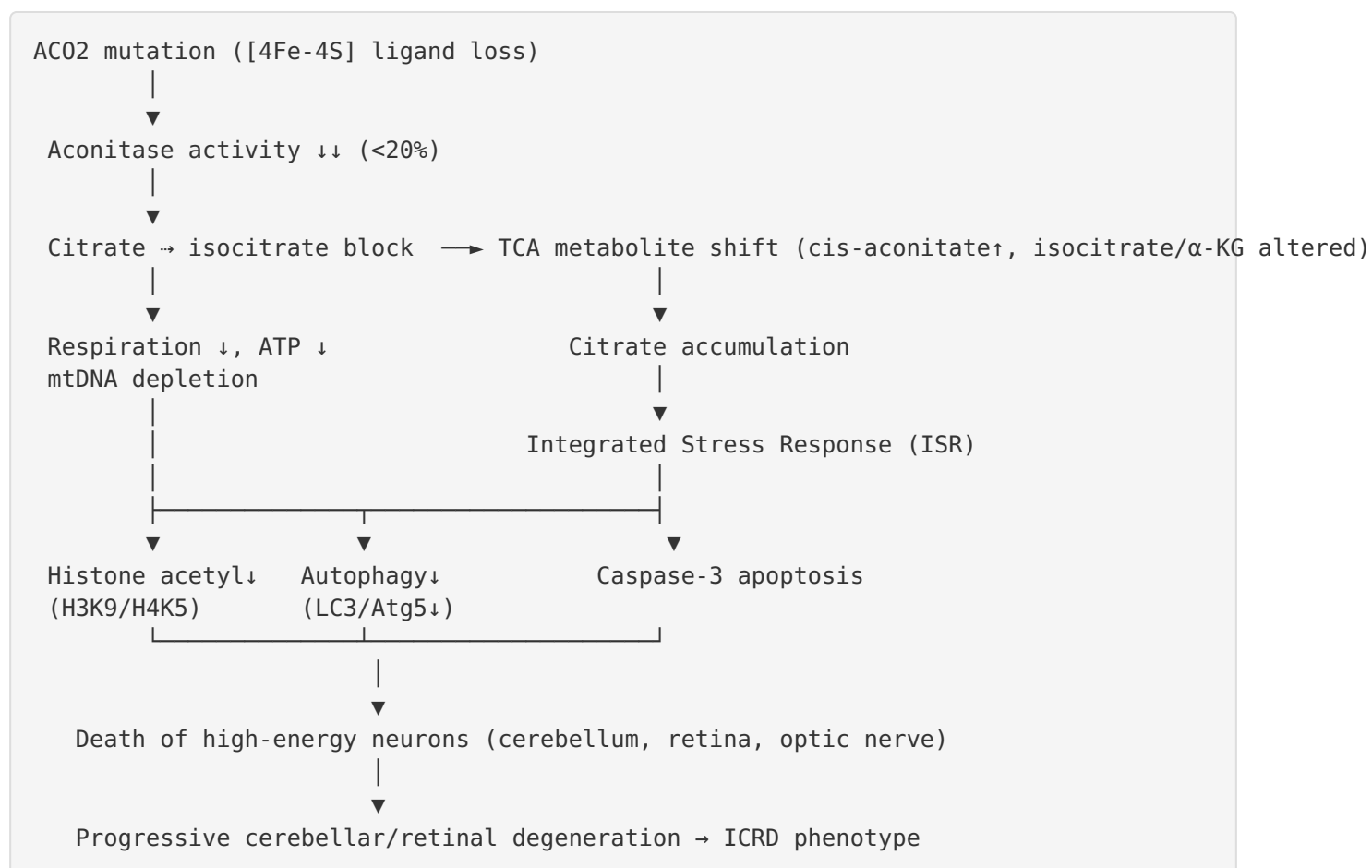
Section 5 — Environmental Information

Not applicable. ICRD is a monogenic recessive disorder with **no environmental, lifestyle, toxic, or infectious contributors**. General mitochondrial stressors (oxidative stress, aminoglycoside-class mitochondrial toxins) are theoretical aggravators of any bioenergetic disorder but have no disease-specific evidence in ICRD. Aconitase is intrinsically redox-sensitive (its [4Fe-4S] cluster is inactivated by superoxide, H₂O₂, NO, ONOO⁻; [PMID: 9171919](#), [PMID: 24266943](#)), so oxidative burden could in principle worsen residual enzyme activity, but this is inferred, not demonstrated in patients.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic ACO2 mutation** damages a substrate-binding residue or a [4Fe-4S] cluster-coordinating cysteine (e.g., Cys448Ser) → **leads to** loss/instability of the catalytic iron-sulfur cluster. (*Demonstrated: enzyme activity <20% of control; F003, F005.*)
2. Cluster/enzyme failure → **results in** inability to convert **citrate** → **cis-aconitate** → **isocitrate** in the TCA cycle. (*Demonstrated: altered plasma cis-aconitate, isocitrate, α -ketoglutarate; F003.*)
3. TCA-cycle block → **leads to** impaired NADH/FADH₂ supply and **bioenergetic (ATP) failure**, with **deficient cellular respiration** and **secondary mitochondrial DNA depletion**. (*Demonstrated in fibroblasts; F003, [PMID: 26992325](#).*)
4. Branch A — **Citrate accumulation** (failure of citrate clearance) → **activates the integrated stress response (ISR)** and impairs cell fitness; reversible by blocking citrate production or promoting citrate efflux. (*Demonstrated in cells and kidney in vivo; F007, [PMID: 41763199](#).*)
5. Branch B — Energy/metabolic stress → **reduced histone acetylation (H3K9, H4K5)** → **downregulation of autophagy genes LC3/Atg5** → impaired autophagic clearance. (*Demonstrated in ACO2-deficient PD model; F007, [PMID: 38007539](#).*)
6. ISR + energy failure + defective autophagy → **caspase-3-mediated apoptosis** of vulnerable cells. (*Demonstrated: Active Caspase-3 up, muscle Aco2-KO; F007, [PMID: 41331265](#).*)
7. Preferential loss of **high-energy-demand, post-mitotic neurons** — cerebellar (Purkinje/ granule) neurons, retinal photoreceptors/ganglion cells, optic nerve axons → **results in** progressive cerebellar atrophy, retinal degeneration, optic atrophy, hypotonia, ataxia, seizures, and developmental regression (the ICRD phenotype). (*Clinical-imaging correlation; F008.*)



Molecular pathways. TCA/Krebs cycle (KEGG hsa00020), oxidative phosphorylation, iron-sulfur cluster biogenesis, integrated stress response. **Cellular processes (GO):** TCA cycle (GO:0006099), aconitate hydratase activity (GO:0003994), 4Fe-4S cluster binding (GO:0051539), generation of precursor metabolites and energy (GO:0006091), autophagy (GO:0006914), apoptotic process (GO:0006915), response to oxidative stress (GO:0006979). **Protein dysfunction:** loss-of-function via cluster destabilization/misassembly. **Metabolic changes:** TCA intermediate flux disruption (diagnostic plasma fingerprint — cis-aconitate, isocitrate, α -KG, phosphoenolpyruvate, hydroxybutyrate; PMID: 28463998); elevated glutamate reported (PMID: 32713659). **Tissue damage:** oxidative-stress-sensitive enzyme, bioenergetic starvation, apoptosis. **Immune involvement:** none primary (mtDNA release can secondarily engage cGAS-STING inflammation via LONP1 axis — inferred, PMID: 42302976).

Subcellular localization (GO Cellular Component): mitochondrion (GO:0005739), mitochondrial matrix (GO:0005759). **Cell types (CL):** cerebellar Purkinje cell (CL:0000121), cerebellar granule cell (CL:0001031), retinal photoreceptor (CL:0000210), retinal ganglion cell (CL:0000740). **Anatomy (UBERON):** cerebellum (UBERON:0002037), retina (UBERON:0000966), optic nerve (UBERON:0000941).

Comparison to related disease. In Friedreich ataxia, frataxin loss secondarily impairs Fe-S enzymes including aconitase, producing mitochondrial iron accumulation and Fe-S enzyme deficiency ([PMID: 9326946](#)) — a mechanistic cousin (Fe-S/aconitase failure) reached by a different primary lesion.

Section 7 — Anatomical Structures Affected

- **Primary organs/systems:** central nervous system (nervous system) — **cerebellum** (predominant), cerebral cortex, optic nerve, and **retina/eye**.
 - **Secondary involvement:** peripheral nerves (neuropathy in longer survivors), skeletal muscle (hypotonia; muscle maturation depends on ACO2 in models).
 - **Tissue level:** nervous tissue (neurons and their axons), retinal neuroepithelium.
 - **Cell populations (CL):** cerebellar Purkinje cells (CL:0000121), granule cells (CL:0001031), retinal photoreceptors (CL:0000210), retinal ganglion cells (CL:0000740), optic nerve axons.
 - **Subcellular (GO CC):** mitochondrion (GO:0005739), mitochondrial matrix (GO:0005759).
 - **Localization / lateralization:** **bilateral, symmetric** CNS atrophy; bilateral optic atrophy and retinal degeneration. UBERON: cerebellum UBERON:0002037, retina UBERON:0000966, optic nerve UBERON:0000941, cerebral cortex UBERON:0000956.
-

Section 8 — Temporal Development

- **Onset:** congenital-to-early-infantile; typical symptomatic onset **2–6 months** of age ([PMID: 22405087](#)); pattern insidious then progressive.
- **Progression:** **progressive neurodegeneration**. MRI is characteristically **normal in the first months**, then develops **progressive global atrophy, predominantly cerebellar** ([PMID: 30689204](#)). Rate is variable and genotype-dependent.
- **Course:** chronic, lifelong, non-remitting. Severe end → death in infancy; moderate → survival into teens (up to 18 years documented) with preserved-but-limited function; mild (OPA9) → normal lifespan.
- **Critical period:** the early-infantile window before irreversible cerebellar/retinal cell loss is the theoretical target for any future disease-modifying (e.g., anaplerotic or gene) therapy.

Section 9 — Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic *ACO2*). The allelic dominant disorder (isolated optic atrophy) exists separately ([PMID: 34056600](#)).
- **Penetrance: effectively complete** for biallelic pathogenic genotypes.
- **Expressivity: highly variable** — an allelic series from OPA9 (mild) to lethal infantile ICRD, largely reflecting residual enzyme activity.
- **Founder effect: yes** — c.336C>G (p.Ser112Arg) recurrent (10/16 cohort; [PMID: 30689204](#)).
- **Consanguinity: contributes** (homozygous founder cases identified by homozygosity mapping).
- **Anticipation / germline mosaicism: not reported / not applicable.**
- **Epidemiology: ultra-rare**; fewer than ~40 patients/families reported worldwide (initial reports noted only ~6 families/5 unique mutations; [PMID: 28545339](#)). No reliable prevalence/incidence estimates; Orphanet lists it as <1/1,000,000.
- **Sex ratio / demographics: no strong sex bias** (autosomal recessive); reported across multiple ethnicities (Middle Eastern founder families, Chinese, European cases).

Section 10 — Diagnostics

Genetic testing (definitive). Diagnosis rests on identifying **biallelic pathogenic *ACO2* variants** by **whole-exome (WES)** or **whole-genome sequencing (WGS)**, or a mitochondrial/cerebellar-ataxia/retinal-dystrophy **gene panel** that includes *ACO2*; targeted testing for the founder c.336C>G in relevant populations. Sanger confirmation and parental segregation establish compound heterozygosity ([PMID: 32713659](#), [PMID: 22405087](#)).

Biochemical / functional tests. Reduced **aconitase enzyme activity** in lymphoblasts/fibroblasts (<20% control) supports pathogenicity; **mtDNA copy number** (depletion) and cellular respiration assays are confirmatory research tools ([PMID: 26992325](#)).

Metabolomic biomarker. A **plasma metabolomic fingerprint** — altered cis-aconitate, isocitrate, α -ketoglutarate, phosphoenolpyruvate, and hydroxybutyrate — serves as a diagnostic signature ([PMID: 28463998](#)).

Imaging. Brain **MRI** — serial imaging shows evolving global atrophy predominantly cerebellar; optic nerve atrophy ([PMID: 30689204](#)). **Ophthalmologic workup** — fundoscopy (optic atrophy, pigmentary retinopathy), visual electrophysiology (ERG/VEP) demonstrating optic atrophy and retinal dysfunction ([PMID: 40210596](#)).

Differential diagnosis. Other infantile cerebellar-atrophy-plus-retinopathy syndromes: neuronal ceroid lipofuscinoses (CLN6, MFSD8/CLN7; [PMID: 39108195](#)), spinocerebellar ataxia type 7 (ATXN7 repeat; [PMID: 37283503](#)), Norrie disease ([PMID: 39965923](#)), Friedreich ataxia, other mitochondrial/Fe-S disorders, and PKAN. ACO2 sequencing plus the TCA metabolite fingerprint distinguishes ICRD.

Screening. Carrier/cascade testing in founder-carrying families; prenatal/preimplantation testing where the familial variants are known. Not part of standard newborn-screening panels.

Section 11 — Outcome / Prognosis

- **Survival:** highly variable by genotype. Severe cases → death in infancy/early childhood; classic cases survive into childhood/teens (up to 18 years reported; [PMID: 22405087](#), [PMID: 28545339](#)); OPA9 → normal lifespan.
 - **Morbidity / function:** severe in classic ICRD — most patients are nonambulatory, nonverbal, cortically/optically visually impaired, with intractable epilepsy in a subset and profound intellectual disability; fully care-dependent.
 - **Complications:** seizures, aspiration/feeding difficulty from hypotonia, contractures, vision loss, and secondary infections typical of severe neurodisability.
 - **Recovery potential:** none — the disease is progressive and neurodegenerative; supportive care is stabilizing at best.
 - **Prognostic factors:** **residual aconitase activity/genotype** is the dominant determinant (missense with partial function → longer survival and partial cognition, e.g., [PMID: 28545339](#)); degree/rate of cerebellar atrophy on serial MRI; seizure burden.
-

Section 12 — Treatment

No disease-specific approved therapy exists. Management is **supportive and multidisciplinary**: antiepileptic drugs for seizures, physical/occupational/speech therapy for hypotonia and developmental support, nutritional support (gastrostomy where needed), low-vision services and ophthalmologic management, and orthopedic care for contractures. (NCIT: supportive care NCIT:C133397; anticonvulsant therapy NCIT:C15229; physical therapy NCIT:C15304.)

Experimental disease-directed therapy — anaplerosis. **Triheptanoin** (an odd-chain C7 triglyceride) was administered to **two brothers with aconitase 2 deficiency** — the first reported disease-directed metabolic intervention for ACO2 deficiency ([PMID: 38668366](#)). Rationale: triheptanoin is metabolized to propionyl-CoA → succinyl-CoA, refilling TCA-cycle intermediates **downstream of the aconitase block** (anaplerosis), theoretically bypassing the citrate→isocitrate lesion. Efficacy remains unproven pending controlled data.

Mechanistically-motivated (preclinical/speculative) targets: promoting mitochondrial citrate efflux or limiting citrate production (reverses ISR/fitness defects in cell/kidney models; [PMID: 41763199](#)); the ACO2-metabolite derivative 4-octyl itaconate rescued mitochondrial dysfunction/apoptosis from ACO2 deficiency in a lung model ([PMID: 41637882](#)); antioxidant strategies given cluster redox sensitivity (theoretical). **Gene therapy/gene replacement** is a rational future direction (WT-ACO2 reintroduction fully rescues the cellular phenotype in vitro; [PMID: 26992325](#)) but not yet clinical.

Pharmacogenomics / immunotherapy / surgery: not applicable as disease-modifying modalities.

Section 13 — Prevention

- **Primary prevention:** not possible for a germline recessive disorder beyond reproductive planning.
- **Genetic counseling & reproductive options:** the mainstay — carrier testing for at-risk couples (especially consanguineous families and founder-allele populations), **prenatal diagnosis** and **preimplantation genetic testing (PGT)** when familial variants are known, and cascade testing of relatives.

- **Secondary/tertiary prevention:** early recognition (metabolite fingerprint + genetics) enables early supportive intervention, seizure control, and complication prevention (aspiration, contractures) — no intervention halts neurodegeneration.
 - **Immunization / public-health / environmental measures:** not applicable.
-

Section 14 — Other Species / Natural Disease

- **Orthologs:** *ACO2* is deeply conserved. Mouse *Aco2* (NCBI Gene 11429); *Drosophila* ortholog *mAcon1*; yeast *ACO1* (functional complementation used to prove human causality, [PMID: 22405087](#)). Bacterial aconitases (*E. coli* AcnA/AcnB) share the [4Fe-4S] cluster architecture with the mammalian mitochondrial enzyme ([PMID: 10585860](#)).
 - **Natural disease in other species:** no well-characterized spontaneous *ACO2*-deficiency disorder in companion animals/wildlife is reported (OMIA); disease knowledge is from engineered models.
 - **Comparative biology:** aconitase's dual role (TCA catalysis + Fe-S/iron sensing) and its redox sensitivity are conserved from bacteria to humans ([PMID: 24266943](#), [PMID: 17205209](#)), making cross-species mechanistic inference robust.
 - **Zoonosis:** not applicable.
-

Section 15 — Model Organisms

Model	Type	Key findings	PMID
Constitutive/skeletal-muscle <i>Aco2</i> knockout mouse	Mammalian, KO	Mice die shortly after birth ; muscle fiber atrophy, disrupted sarcomeres, increased Active Caspase-3 (apoptosis); aconitase essential for muscle maturation	41331265
<i>ACO2</i> A252T knock-in mouse	Mammalian, KI	Aggravated dopaminergic neurodegeneration; downregulated autophagy (LC3, Atg5) via reduced H3K9/H4K5 histone acetylation	38007539
<i>Drosophila</i> (<i>mAcon1</i>)	Invertebrate	Pan-neuronal knockdown/overexpression reduces longevity, locomotion, activity; disrupts sleep/circadian rhythm; eye mis-expression → impaired visual synaptic transmission and neurodegeneration — mirrors human ICRD	40210596
Patient fibroblasts	In vitro (cellular)	Aconitase activity <20%, deficient respiration, mtDNA depletion; fully rescued by WT-ACO2 reintroduction	26992325
Yeast (<i>ACO1Δ</i>) complementation	Cellular	Mutant human ACO2 fails to rescue aconitase-null yeast — proves loss of function	22405087

Phenotype recapitulation: the *Drosophila* model recapitulates neuronal dysfunction, visual/retinal degeneration, and locomotor decline; the knock-in mouse recapitulates the metabolism→epigenetics→autophagy→neurodegeneration axis. **Limitations:** constitutive KO is lethal (limiting adult CNS study without conditional alleles), and no model perfectly reproduces the full human cerebellar-retinal-cognitive triad.

Key Findings (with statistical evidence)

F001 — ICRD is caused by biallelic ACO2 variants. Homozygosity mapping + WES in 8 individuals from 2 families identified homozygous p.Ser112Arg (c.336C>G); patient lymphoblast aconitase activity was severely reduced, and mutant human ACO2 failed to complement a yeast *ACO1* deletion. *"Homozygosity mapping followed by whole-exome sequencing disclosed a Ser112Arg mutation in ACO2, encoding mitochondrial aconitase... Specific aconitase activity in the individuals' lymphoblasts was severely reduced"* (PMID: 22405087). A 16-patient cohort confirmed biallelic pathogenic variants (PMID: 30689204).

F002 — Phenotypic spectrum from OPA9 to severe ICRD. *"Biallelic variants in ACO2 are purported to cause two distinct disorders: infantile cerebellar-retinal degeneration (ICRD)... and optic atrophy 9 (OPA9), characterized by isolated ophthalmologic phenotypes"* (PMID: 32449285). Dominant ACO2 variants also cause isolated optic atrophy (PMID: 34056600).

F003 — Metabolomic fingerprint and cellular energy defects. *"...metabolites with affected plasma concentrations including the tricarboxylic acid cycle metabolites cis-aconitate, isocitrate and alpha-ketoglutarate, as well as phosphoenolpyruvate and hydroxybutyrate"* (PMID: 28463998). *"ACO2 enzyme activity was <20% of that observed in control cells... deficiency in cellular respiration and, for the first time,... mitochondrial DNA depletion"* — rescued by gene reintroduction (PMID: 26992325).

F004 — Strong gnomAD constraint. ACO2 (ENSG00000100412): pLI 0.97, LOEUF 0.495 (o/e LoF 0.367), missense-Z 4.58, LoF-Z 4.94 — an essential, constraint-heavy gene, consistent with recessive ICRD and dominant optic-atrophy mechanisms.

F005 — Variants strike catalytic Fe-S residues. UniProt Q99798 annotates [4Fe-4S] ligands Cys385/448/451; p.Cys448Ser removes a ligand and p.Met393Ile lies adjacent to Cys385 (PMID: 32713659).

F006 — Triheptanoin anaplerotic therapy trialed. First disease-directed metabolic intervention: *"Anaplerotic Therapy Using Triheptanoin in Two Brothers Suffering from Aconitase 2 Deficiency"* (PMID: 38668366).

F007 — Citrate accumulation → ISR → apoptosis; KO lethal. *"Disrupting citrate catabolism activates the integrated stress response and impairs cell fitness... reversed by preventing citrate production or promoting mitochondrial citrate efflux. In vivo, ACO2 deficiency induces citrate accumulation and triggers tubular degeneration in the kidney"* (PMID: 41763199). Muscle

Aco2-KO mice "died shortly after birth" with caspase-3 apoptosis (PMID: 41331265). "...autophagy-related genes LC3 and Atg5 was significantly downregulated via inhibited histone acetylation at the H3K9 and H4K5 sites" (PMID: 38007539).

F008 — Characteristic clinical/MRI course. *"Most patients present in early infancy with severe truncal hypotonia, truncal ataxia, variable seizures, evolving microcephaly, and ophthalmological abnormalities of which the most dominant are esotropia and optic atrophy with later development of retinal dystrophy" and "Brain MRI is typically normal within the first months but global atrophy gradually develops affecting predominantly the cerebellum" (PMID: 30689204).*

Mechanistic Model / Interpretation

ICRD is best understood as a **primary bioenergetic + citrate-toxicity disorder** of neurons. The single enzymatic lesion (aconitase failure) produces two converging insults: (i) an **energy-supply deficit** (TCA flux ↓, respiration ↓, mtDNA depletion), and (ii) a **toxic-substrate accumulation** (citrate build-up) that actively triggers the integrated stress response, apoptosis, and epigenetically-mediated autophagy suppression. These converge on the death of post-mitotic, high-energy-demand cells — cerebellar neurons, retinal photoreceptors and ganglion cells, and optic nerve axons — explaining the disease's signature cerebellar-retinal predilection. The **allelic series** (OPA9 ↔ ICRD) is naturally explained by a residual-activity model: variants with partial function spare the CNS and manifest only in the exquisitely oxidative-metabolism-dependent optic system, while near-null biallelic genotypes produce lethal multisystem disease. The finding that citrate efflux, blocking citrate production, or 4-octyl itaconate can reverse cellular phenotypes, and that WT-ACO2 fully rescues fibroblasts, identifies **actionable, druggable nodes downstream of the mutation**.

Evidence Base

PMID	Contribution	Source type
22405087	Establishes <i>ACO2</i> causality; founder variant; yeast complementation	Human + in vitro
30689204	Largest cohort (n=16); clinical/MRI delineation; founder frequency	Human clinical
32449285	ICRD↔OPA9 spectrum	Human clinical
34056600	Dominant <i>ACO2</i> optic atrophy	Human clinical
26992325	Enzyme <20%, respiration defect, mtDNA depletion, gene rescue	In vitro
28463998	Plasma metabolomic diagnostic fingerprint	Human biomarker
32713659	Cys448Ser / Met393Ile — Fe-S cluster structural basis	Human + structural
28545339	Moderate phenotype, preserved cognition, longer survival	Human clinical
41763199	Citrate clearance → ISR → cell fitness (2026)	In vitro + mouse
41331265	<i>Aco2</i> KO lethal; caspase-3 apoptosis (2025)	Mouse
38007539	Metabolism→histone-acetylation→autophagy axis	Mouse + fly
40210596	<i>Drosophila</i> ICRD model recapitulating neuro/visual phenotypes	Invertebrate
38668366	Triheptanoin anaplerotic therapy	Human (n=2)
9326946	Aconitase/Fe-S deficiency in Friedreich ataxia (comparator)	Human

Limitations and Knowledge Gaps

- **Ultra-rarity:** Total reported patients number in the low dozens; no prevalence/incidence, no natural-history registry, no formal QoL data.
- **Genotype–phenotype correlations** are inferred from small cohorts; the residual-activity model is plausible but not quantitatively validated across the allelic series.

- **No CNS-specific animal model** fully recapitulates the human cerebellar-retinal-cognitive triad; constitutive KO lethality limits adult neuro-study without conditional alleles.
- **Therapeutics unproven:** triheptanoin evidence is anecdotal (n=2, no controls); downstream targets (citrate efflux, 4-OI, antioxidants) are preclinical only; no gene therapy in trials.
- **Mechanistic branches** (ISR, epigenetic-autophagy, apoptosis) are largely demonstrated in non-neuronal or non-human systems and inferred for the patient CNS.
- Some causal steps (oxidative-stress aggravation, immune/cGAS-STING involvement) remain **inferred, not demonstrated**, in ICRD patients.

Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC cerebellar and retinal organoids** carrying defined *ACO2* alleles to directly test the citrate-toxicity/ISR/apoptosis chain in the affected human cell types and to correlate residual enzyme activity with degeneration rate.
 2. **Conditional (neuron- and photoreceptor-specific) *Aco2* knockout / knock-in mice** to bypass embryonic lethality and model CNS/retinal disease longitudinally.
 3. **Controlled anaplerosis trial:** formal evaluation of triheptanoin (and comparison with citrate-efflux promotion or 4-octyl itaconate) using the plasma TCA-metabolite fingerprint as a pharmacodynamic biomarker.
 4. **AAV-mediated *ACO2* gene replacement** proof-of-concept in models, leveraging the demonstrated full cellular rescue by WT-*ACO2*.
 5. **Prospective natural-history study and registry** with standardized serial MRI (cerebellar volumetrics), ERG/VEP, and developmental metrics to define progression and trial endpoints.
 6. **Structure-guided variant classification:** map all reported variants onto the aconitase [4Fe-4S] and substrate-binding architecture (Q99798) and correlate predicted structural impact with clinical severity to formalize the residual-activity model.
-

Report compiled from 8 confirmed findings and 32 reviewed papers across 5 investigation iterations. Evidence types are distinguished as human clinical, model organism, in vitro/cellular, and computational/structural throughout.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery